Jainam Karania

AI/ML & Data Enthusiast | Frontend Engineer
Bengaluru, Karnataka, India | +91-9104955662 | jainamkarania05@gmail.com
Portfolio : https://jainam-karania.vercel.app/
/JainamKarania @JainamKarania

SKILLS

- **Programming Languages:** Python, JavaScript/TypeScript.
- **ML & Data Libraries:** NumPy, Pandas, scikit-learn, data handling, data cleaning, data visualization, feature engineering & selection, train/test splits, cross-validation.
- **Containerization:** Docker.
- **Model evaluation metrics:** accuracy, precision, recall, F1, RMSE.
- **Generative AI & NLP:** LLM integration, prompt engineering, RAG pipelines, Lang Chain.
- **Databases:** PostgreSQL, MySQL, relational schema design.
- **Productivity & Collaboration Tools:** Google Workspace (Docs, Sheets, Slides, Drive), Git.
- **Frontend :** React, Web flow, Material UI, Jest, Tailwind CSS, Bootstrap, GSAP, HTML, CSS.
- **Backend:** Node, JavaScript, TypeScript, SQL, Advanced SQL.
- **AI tools :** ChatGPT, Claude, Gemini and Copilot.
- **Cloud :** Google Cloud Platform, AWS.

EXPERINCE

AI/ML Upskilling & Independent Projects *Feb 2025 - Present*

- Built a foundation in Python, Machine Learning, and Generative AI through structured learning and hands-on projects.
- Explored LLMs, RAG, embeddings, prompt engineering, and Gemini APIs by developing practical AI applications.
- Developed AI-backed applications using FastAPI, React, Python, and third-party AI APIs, combining frontend experience with backend and AI development.

Frontend Engineer | Pure Connectt Technologies, Ankleshwar *Oct 2024 – Feb 2025*

- Explored and integrated AI-assisted features into application workflows, reflecting a self-driven interest in applying AI tooling beyond standard frontend development.
- Built and debugged UI components in React with TypeScript, developing a habit of systematic testing and quality checks - a discipline carried directly into current ML project evaluation work.
- Collaborated with a cross-functional team to translate requirements into working features, strengthening the ability to scope and iterate on technical problems independently.

Webflow Developer | TheCSSAgency, Surat *Oct 2023 – Feb 2024*

- Took ownership of translating Figma designs into production-ready, responsive builds, learning new tools and platforms independently.
- Collaborated directly with designers to troubleshoot issues and incorporate feedback experience in structured problem-solving that transfers to debugging data and model pipelines.
- Followed structured naming and architecture conventions to keep projects organized, maintainable, and easy for others to pick up.

COURSES & CERTIFICATIONS

GenAI Exchange Program powered by Google Cloud & Hack2Skill | [View Certificate]

- Ranked 14 out of 2 lakhs+ candidates enrolled in this program, completed 5 GenAI courses provided by Google via Google Skills Boost and Hack2skill.

Forward Program | McKinsey & Company | [View Certificate]

- Gained exposure to McKinsey's approach to strategic thinking, which sharpened how I frame and tackle ambiguous, real-world problems.

Machine Learning with Python: Foundations – LinkedIn Learning | [View Certificate]

- Built a solid grounding in ML fundamentals using Python, covering the core algorithms and concepts that most models are built on.

Machine Learning and AI Foundations: Clustering and Association - LinkedIn Learning | [View Certificate]

- Explored unsupervised learning techniques, learning how algorithms group data and uncover hidden patterns without labelled outcomes and learned to apply clustering and association rule mining to real datasets, useful for tasks like customer segmentation and market basket analysis.

Data Science Foundations: Data Mining - LinkedIn Learning | [View Certificate]

- Learned the process of digging through raw data to extract meaningful patterns and insights that actually drive decisions and practiced core data mining techniques and workflows, building a stronger intuition for how to approach messy, real-world datasets.

Data Science on Google Cloud Platform: Designing Data Warehouses | [View Certificate]

- Learned how to design and structure data warehouses on GCP, understanding how large-scale data systems are built for analytics and gained practical knowledge of cloud-based data architecture, which helps in thinking about scalability and performance from day one.

PROJECTS

Customer Churn Prediction | [Repo Link] *Aug 2026 to Present*

- Engineered and compared 4 classification models - Logistic Regression, Random Forest, XGBoost, Neural Network on imbalanced telecom churn data 27% positive class, achieving F1=0.644 and ROC-AUC=0.849 with the best-performing model.
- Identified and fixed a silent data visualization bug and a false-ordinality encoding issue Label Encoder on nominal categories, improving model input quality prior to training.
- Tuned Random Forest via RandomizedSearchCV explicitly optimizing for Recall, surfacing a precision-recall trade-off that informed model selection based on business cost of missed churners vs. false alarms and delivered a documented, reproducible comparison showing the simplest model Logistic Regression outperformed three more complex alternatives on F1 and ROC-AUC - a finding validated through cross-validation, not a single train/test split.

Retail Customer Segmentation via RFM + Dimensionality Reduction | [Repo Link] *Aug 2026 to Present*

- Cleaned and transformed 541,909 raw e-commerce transaction records into a customer-level dataset 4,338 customers via RFM (Recency, Frequency, Monetary) feature engineering, handling cancellations, returns, and missing customer IDs.
- Applied PCA to reduce 5 engineered features to 2 interpretable components explaining 90% variance in 3 dimensions, identifying a primary "customer engagement/value" axis and a secondary "purchase pattern" axis.
- Diagnosed and corrected a feature-skew bug that had caused one principal component to be dominated by a single outlier-heavy feature, then validated the fix by tracing model output back to individual customer records and compared PCA against t-SNE for non-linear structure discovery, and applied K-Means/Hierarchical/DBSCAN clustering, correctly diagnosing DBSCAN's failure mode via direct density measurement rather than accepting a visual mismatch at face value.

Financial Report Analysis | [Repo Link] *Jan 2026 – Present*

- Built an NLP pipeline to extract text from financial report PDFs and identify key metrics like Revenue, Net Income, EPS, Operating Margin using pattern-based extraction, validated with a custom precision/recall/F1 evaluation framework.
- Fine-tuned a T5 transformer model for financial report summarization using a proper train/validation/test split and evaluated outputs with ROUGE metrics instead of manual inspection.
- Expanded and diversified the training dataset to stress-test extraction accuracy against varied phrasing and incomplete data, improving robustness over an initial baseline version.

EDUCATION QUALIFICATIONS

- MCA | Jain University, Bangalore *CGPA: 7.43 (Sep 2021 to May 2023)*
- BCA | Shri Manilal Kadakia College, Ankleshwar *CGPA: 7.20 (Aug 2018 to May 2021)*